Design review - MelansonTech Isolated Shunt Amplifier, Rev 3.1

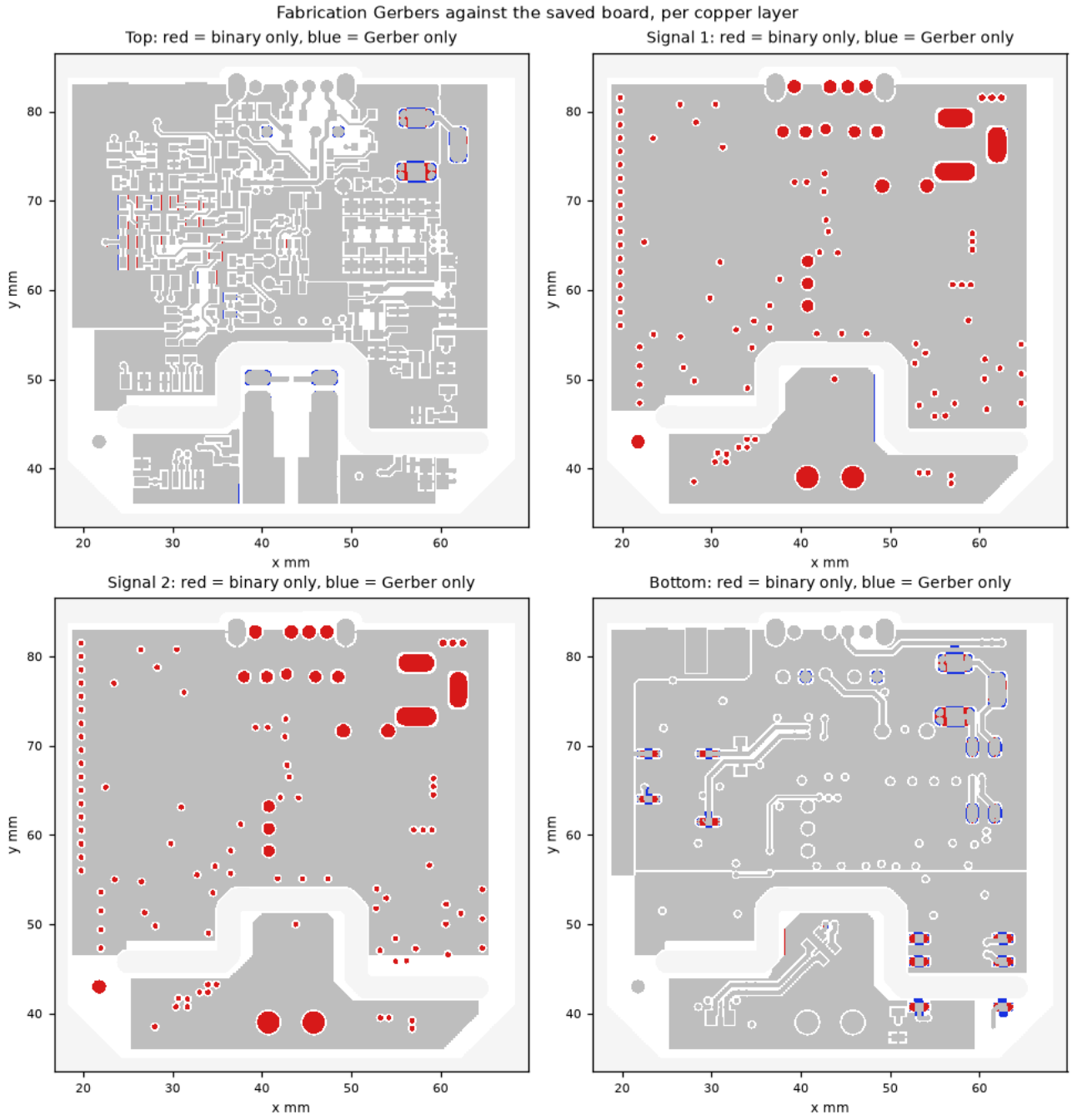
Review date: 2026-10-29. **Generated** from `hardware/review/*.json` by `make_review.py`; every number below is read from a result file and marked **simulated**, **calculated** or **checked**.

Board state reviewed	
board	MelansonTech Isolated Shunt Amplifier Rev3.CSPcbDoc, saved 2026-08-20 20:37 (PDT)
sheets	Rev3 amplifier sheet 20:35, Rev3 PSU sheet 20:34, same day
outputs	Default Configuration/Outputs, written 2026-08-20 20:43
commit	512311d (2026-09-09), repository MelansonTech/Melanson-Tech-Isolated-Shunt-Amplifier, clone of 2026-10-28
bench	one Rev 3.1 board characterised 2026-09-05/06 (docs/TEST_REPORT.md)

Scope: the five gates that run on every review, then every area of the catalogue (isolation, power and rails, thermal, analog bias / loops / noise / bandwidth, EMC, DFM and the fabrication package, enclosure, sourcing and lifecycle, documentation). The schematic-level review of 2026-08-20 (`docs/DESIGN_REVIEW.md`, findings F1-F15) is carried in the register and re-examined where this run reaches further.

Gates

Gate	Verdict	What it found	Script
Outputs vs saved board	PASS	top 2.9 mm2 / bottom 11.7 mm2 of residual one-sided difference (distributed raster noise); inner layers differ only by the 68.5 mm2 of unused inner pads CircuitStudio removes from the Gerbers	gerber_vs_binary.py
Placement vs pick-and-place	PASS	94 of 95 parts at their reference point; the P&P names R10 where the board's designator is blank (sync-2)	pnp_vs_board.py
BOM vs board	PASS*	95 designators each way; the only miss is the unlinked R10 (sync-2); four comments differ (bom-1, bom-2)	bom_vs_board.py, comment_diff.py
Sheet vs board netlist	FAIL	5 pins differ, all one part: R21 reversed and renamed 0 R on the sheet after the last ECO (sync-1); the committed sheets match the disk; at 053f1d7 the board matched	netlist_diff.py
Connectivity vs function	PASS	11 interfaces, 42 pins, every pin reaches its target on sheet, board and HEAD except R28's wiper, which fails on the sheet only through the R21 reversal	connectivity_check.py
Pin constraints vs sheet	n/a	no programmable device	
Port directions	n/a	two flat sheets joined by power ports and one port (HighGain); nothing to check	port_directions.py
Findings register	carried	53 entries, 49 open	register.py



Fabrication Gerbers against the saved board, per copper layer (checked). Red is copper only in the binary, blue only in the Gerbers; the inner-layer red is the unused inner pads of vias and through-hole parts, which the output job strips.

Open findings, by severity

Id	Sev	Area	Finding
F4	1	legacy	R28 is 100 ohm on the schematic and CT94EW103 (10 k) in the BOM
sourcing-1	1	sourcing	LM7321MF/NOPB (U5, U9): OBSOLETE
F11	2	legacy	No reverse-polarity, fuse or TVS on the 15 V input; D6 clamp topology
F12	2	legacy	SW1 carries the whole board supply at its 100 mA / 12 V rating
F13	2	legacy	BOM lines C8, C43, R23 are generic placeholders
F15	2	legacy	Rev 3.1 silkscreen legends disagree with the schematic
F2	2	legacy	+12 rail unregulated below ~14.9 V at the jack
F5	2	legacy	U6 HT7550-1 dissipates ~0.55-0.6 W in SOT-89
analog-4	2	analog	LM7321 bias current (0.4 uA typ, up to 2 uA) through the R21 asymmetry: 4 mV typ / 20 mV max offset between the two references with the board's 10 k, zero with the sheet's 0 R
bom-1	2	bom	D6: sheet/BOM says BAS31, board says BAT54SLT1G
bom-2	2	bom	R2: sheet/BOM says ERA-6VRW8201V, board says RR1220P-8251-D-M
bom-3	2	bom	R9: sheet/BOM says ERA-6VRW8201V, board says RR1220P-8251-D-M
dfm-1	2	dfm	Pours come within 0.21 mm of the barrier slot wall on every layer
dfm-2	2	dfm	Barrier slot is drawn on the Outline layer as a 2.5 mm stroked path, not a closed cut-out contour
dfm-3	2	dfm	A .GKO (keep-out) file with an 11.5 x 3.9 mm rectangle is in the fabrication set
emc-1	2	emc	Conducted emission at the 15 V port from PSU1: worst margin -4.8 dB to CISPR 32 Class B AVG (10-30 MHz), module scan de-embedded
isolation-1	2	isolation	Barrier creepage 3.43 mm (bottom) / 3.68 mm (top) around the slot's left end, below the 4.0 mm basic figure for 400 V
isolation-2	2	isolation	Aluminium case rail (bonded to ISenseRTN) is 3.60 mm from PSU1's -Vout pad along the bottom surface at 1 mm groove engagement
isolation-3	2	isolation	Free 3.5 mm plated hole at (21.75, 43.0) sits 2.1 mm from measured-side copper and 1.8 mm from output-side copper
isolation-4	2	isolation	Pours on every layer come within 0.21 mm of the slot wall
isolation-5	2	isolation	No design rule enforces the barrier
power-1	2	power	x100 full scale (5.0 V) is at the output clamp at 13 V in: two Schottky drops plus the regulator dropout
power-2	2	power	U6 (HT7550-1, SOT-89) dissipates 0.62 W at 15 V and 0.71 W at 16 V against a 0.5 W package rating
power-3	2	power	SW1 carries 93-97 % of its 0.1 A rating at 15 V (rated 12 V) and makes/breaks the board supply
sourcing-2	2	sourcing	G6L-1F DC12 (Rel1): conflicting
sourcing-3	2	sourcing	PJ-002A (J4): conflicting
sourcing-4	2	sourcing	DR331-105BE (CM1): conflicting
sourcing-5	2	sourcing	CL31B475KBHNNNE (C11, C12): NRND (LCSC)

Id	Sev	Area	Finding
sync-1	2	sync	Sheet and board differ: R21 is fitted the other way round and renamed on the sheet (0 ohm link) after the last ECO
sync-2	2	sync	R10 exists on the sheets but the board's R10 footprint is not linked to it
thermal-1	2	thermal	The slot makes the shunt area a thermal island: R5 at 10 A reaches 110 C in free air and 138 C in the case at 40 C with 2.5 mm2 leads as fins
F10	3	legacy	Difference-amp CMRR relies on four discrete resistors
F14	3	legacy	Relay contact resistance sits in the x100 gain network
F3	3	legacy	R21 asymmetry between the two reference buffers (10 k vs 0)
F9	3	legacy	C16, C17, C28 at 3 pF are below the layout parasitics
analog-1	3	analog	U9_x100 loop: phase margin 26 deg (simulated, LM7321 model at its 10 V group)
analog-2	3	analog	x100 bandwidth is set by C9 (47 nF) against the 50 ohm output resistors: simulated 58 kHz, measured 60 kHz
analog-3	3	analog	Input-referred noise 1.7 mA rms (x100, 60 kHz) and 5.4 mA rms (x10, 580 kHz), dominated by the Si8921
analog-5	3	analog	D6 as drawn is an antiparallel clamp across AIP/AIN (+/-0.7 V with BAS31), not a series pair
dfm-4	3	dfm	No fiducials and no test points on the board
documentation-1	3	documentation	Documents give the board as 68.25 x 85.00 mm; it is 50.00 x 50.00 mm
documentation-2	3	documentation	README says Revision 3 'has not been fabricated or tested' while the test report characterises a Rev 3.1 board
documentation-3	3	documentation	The board file's stackup is not a fabricated one: 1.524 mm between top and the first plane, then 0.254 / 0.254
documentation-4	3	documentation	C7 is a 10 x 16 mm can (A750 MV); upright it cannot fit the 24 mm case, lying flat it clears by ~3 mm with 1.5 mm walls
documentation-5	3	documentation	Rev 1 Bode files and the sheet note 'Circuit Requires 300mW' are stale
isolation-6	3	isolation	J2's 400 V rating applies to overvoltage category II / pollution degree 2 only
power-4	3	power	Relay coil at 16 V is 130 % of rating, within the G6L's 150 % maximum to 70 C
sourcing-6	3	sourcing	Part numbers that no distributor listing matched: NXJ1S0505T6MC-R7, ERA-6VRW8201V, ERA-6AEB471V, C0805X102J1GACAUTO
thermal-2	3	thermal	J2-to-shunt pours rise ~21 C at 10 A (IPC-2221, 1 oz, top layer only)

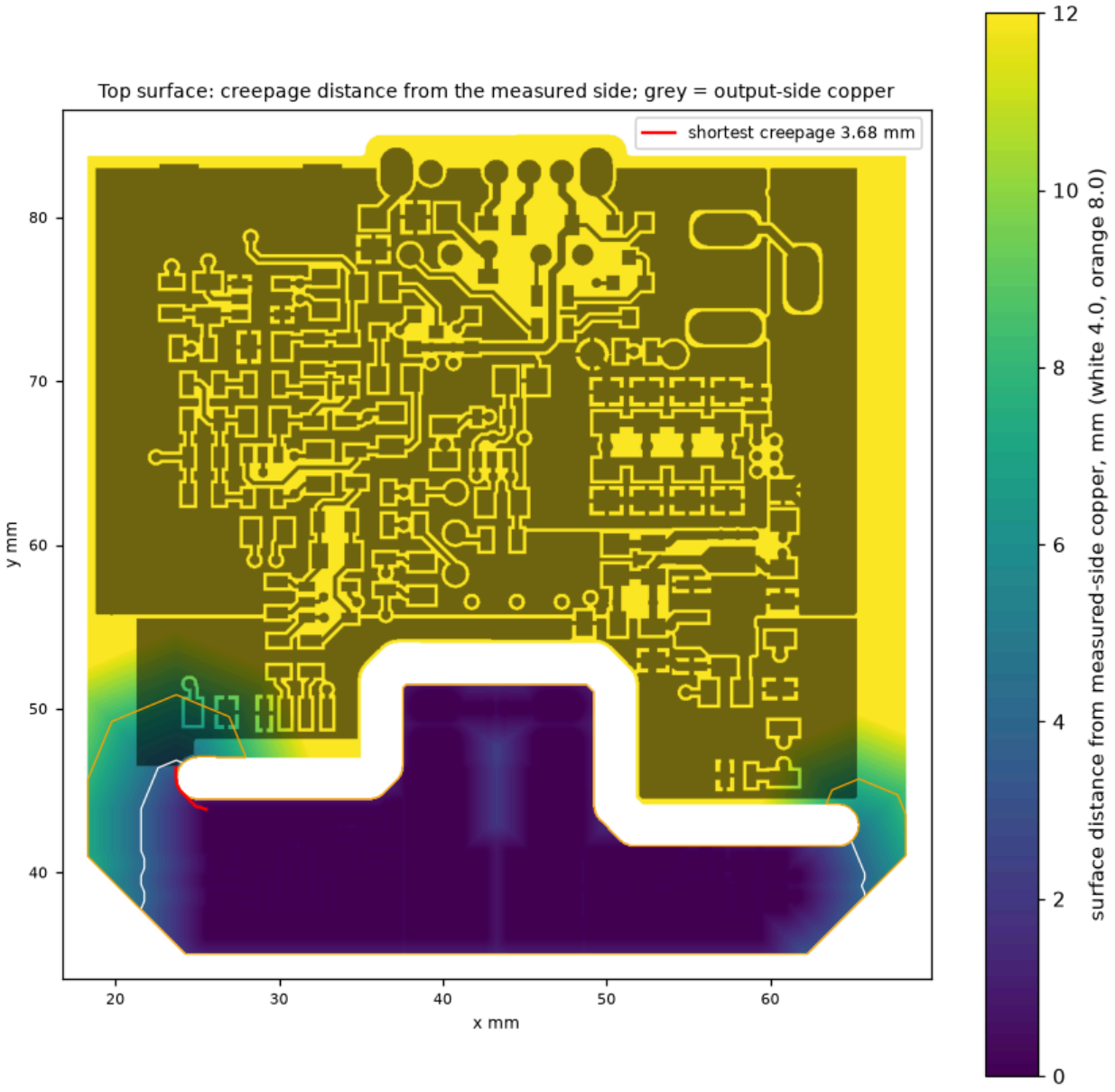
Isolation and safety

Checked and simulated (geometry from the board file, 0.05 mm raster; creepage as a geodesic on the surface inside the outline and around the slot). Standard figures are IEC 60664-1 and are assumptions until a clause is chosen: creepage at 400 V, pollution degree 2, material group IIIb = **4.0 mm basic / 8.0 mm**

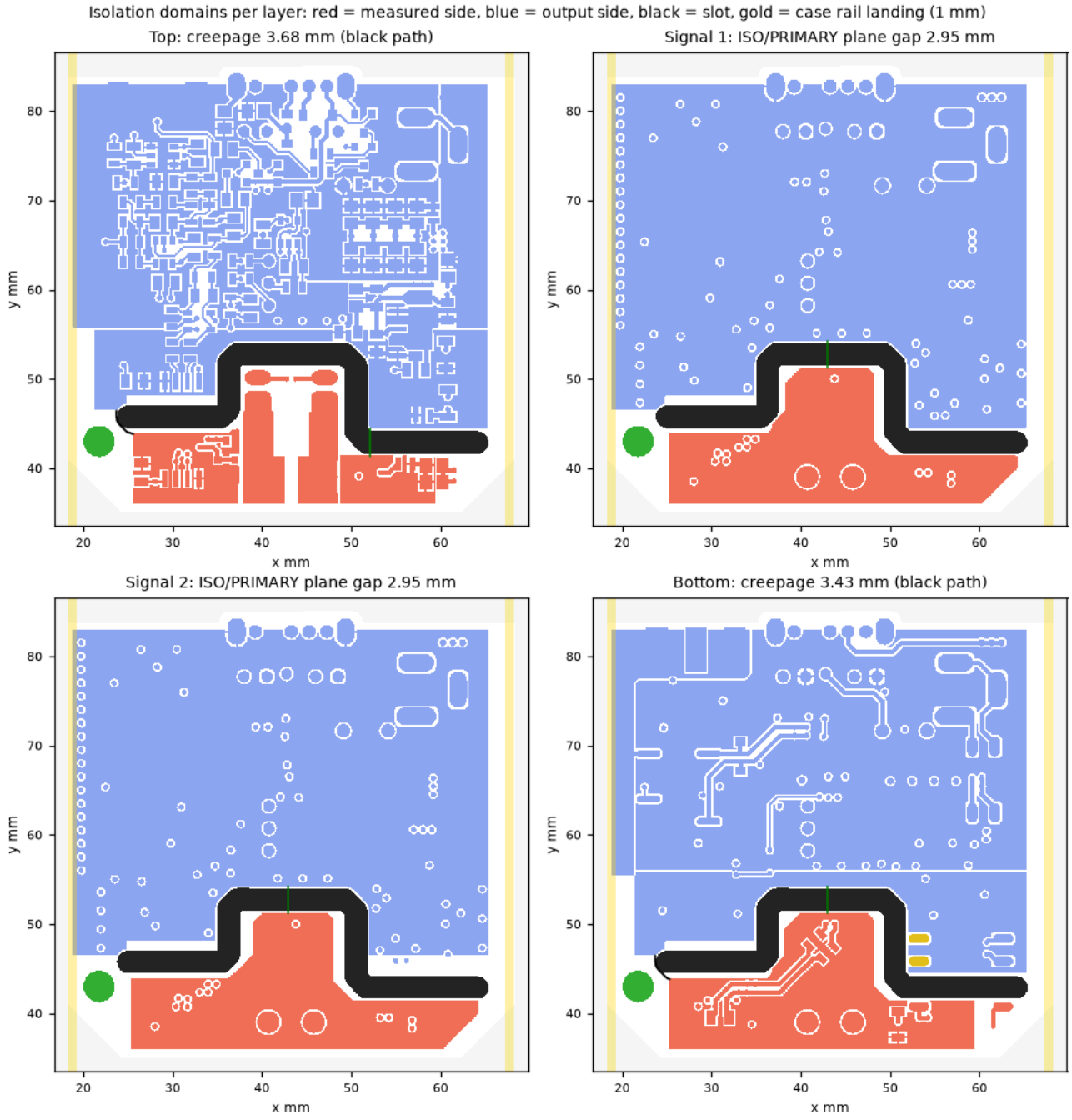
reinforced (5.0 mm at 500 V); clearance 1.5 mm for a 2.5 kV impulse, 3.0 mm for 4 kV. The DC-DC module is UL 62368-1 recognised to 400 Vrms basic / 200 Vrms reinforced with 4.0 mm of its own; the Si8921 wide-body package has 8.5 mm.

Layer	Measured-side copper mm2	Output-side copper mm2	Straight gap in plane mm	Surface creepage mm	Copper to slot wall mm	Copper to outline mm
Top	244.7	1246.0	3.100	3.680	0.212	0.550
Signal 1	326.1	1521.2	2.950	embedded	0.212	0.500
Signal 2	321.6	1498.4	2.950	embedded	0.212	0.550
Bottom	293.8	1431.8	2.950	3.430	0.212	0.550

The shortest surface path, 3.43 mm on the bottom and 3.68 mm on the top, runs from the GND_ISO / +5_ISO pours at [25.3, 43.86] round the slot's left end to the output-side +5 pour at [23.7, 46.51], which wraps 2.5 mm past the slot tip into the bridge. The 2.5 mm slot itself gives an in-plane air gap of 3.10 mm, which clears 400 V; the creepage does not.



Top surface: distance along the surface from measured-side copper (simulated). The white contour is 4.0 mm, orange 8.0 mm; the red path is the shortest creepage.



The two domains per layer with the slot (black), the shortest paths and a 1 mm case-rail landing (gold).

Rail engagement mm	Creepage to the rail, bottom mm	Creepage to the rail, top mm	Straight line to the rail edge, bottom mm
0.5	4.10	6.32	4.10
1.0	3.60	5.61	3.60
1.5	3.10	5.05	3.10
2.0	2.60	4.55	2.60

The aluminium case is bonded to ISenseRTN through the SMA nut, so its rails are output-side metal. The right-hand rail lands beside PSU1's -Vout pad: [63.7, 40.56] to [67.3, 40.56] on the bottom surface at 1 mm engagement. The free 3.5 mm plated hole at (21.75, 43.0) is 2.1 mm from measured-side copper and 1.8 mm from output-side copper on the top layer; PSU1's NC pins 12/14 are on the output side, 0.25 mm from its copper, which is fine because the datasheet says they are not connected internally.

Power budget, rails and headroom

Calculated, sources named per line (`power_budget.py`): the module's input current is the residual of the measured 95.8 mA total, which implies 43 % module efficiency at its 137 mW load.

Load	Rail	mA	Source	Basis
U3	+12	1.00	DS	OPA197 IQ 1.0 mA typ
U8	+12	1.00	DS	OPA197 IQ 1.0 mA typ
U5	+12	1.35	DS	LM7321 IS 1.35 mA typ
U9	+12	1.35	DS	LM7321 IS 1.35 mA typ
R22/R17	+12	0.06	CALC	100k + 100k reference divider
R26/R28/R29	+12	0.06	CALC	100k + 100 R + 100k trim divider
D4 or D5	+12	0.88	CALC	one blue LED through 10k, Vf 3.2 V (BOM description)
R13	+12	0.60	CALC	10k from U3 out (~Vref) to GND, sourced by U3
R15	+12	0.60	CALC	10k from U8 out (~Vref) to GND, sourced by U8
U1	VregIn	3.84	DS	UA78L12 bias 3.84 mA typ (new chip)
U2 VDDB	+5	3.50	DS	Si8921 output side 3.5 mA typ
PSU1 in	+5	63.26	MEAS	residual of the measured 95.8 mA total at 15 V, x100
R6	+5_ISO	22.73	CALC	220 R preload on the isolated 5 V
U2 VDDA	+5_ISO	4.60	DS	Si8921 input side 4.6 mA typ
U6	VregIn	0.00	DS	HT7550-1 quiescent 2.5 uA
Rel1 coil	HighGain	18.30	CALC	G6L-1F DC12 coil 800 R at HighGain

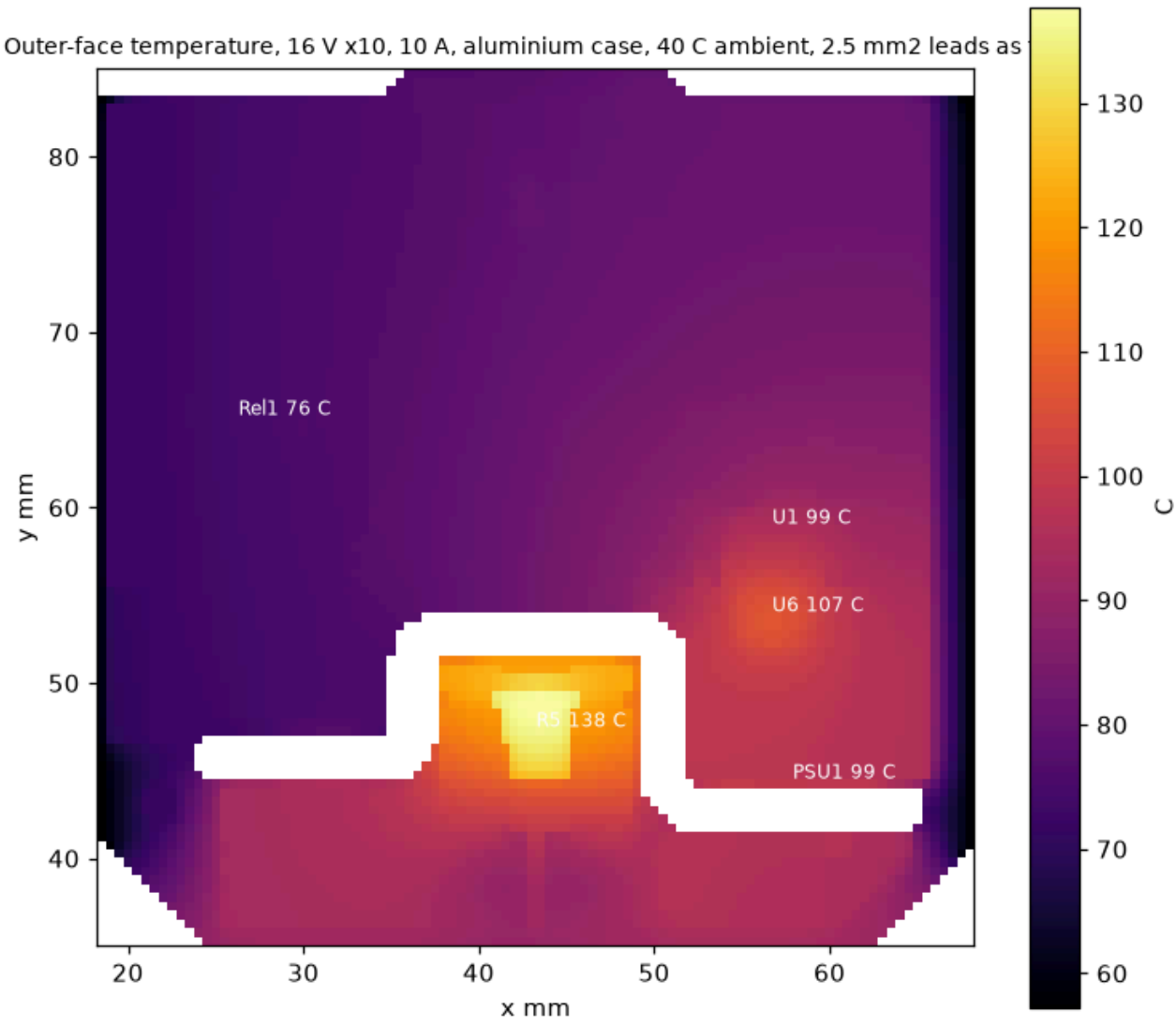
Vin V	Range	VregIn V	+12 typ V	+12 guaranteed V	Headroom typ V	Headroom guaranteed V	Coil %	U6 W	SW1 % of 0.1 A
13.0	x10	12.64	10.94	10.14	+4.17	+3.77	0.0	0.51	77.3
13.0	x100	12.28	10.58	9.78	-0.01	-0.41	105.3	0.49	93.0
14.0	x10	13.64	11.94	11.14	+4.67	+4.27	0.0	0.58	77.5
14.0	x100	13.28	11.58	10.78	+0.49	+0.09	113.7	0.55	94.5
15.0	x10	14.64	12.00	12.00	+4.70	+4.70	0.0	0.64	77.5
15.0	x100	14.28	12.00	11.78	+0.70	+0.59	122.0	0.62	95.8
15.6	x10	15.24	12.00	12.00	+4.70	+4.70	0.0	0.68	77.5
15.6	x100	14.88	12.00	12.00	+0.70	+0.70	127.0	0.66	96.6
16.0	x10	15.64	12.00	12.00	+4.70	+4.70	0.0	0.71	77.5
16.0	x100	15.28	12.00	12.00	+0.70	+0.70	130.3	0.69	97.1

Headroom is the output clamp (+12 less the 0.30 V the OPA197 output loses at its ~5 mA load, the figure the bench measured as 5.70 V above a 6.0 V reference) less the reference less full scale (5.0 V on x100, 1.0 V on x10). SW1's pole is terminal 2 (C&K: 1-2, 2-3, 2-4), so the x100 position feeds VregIn through D1 as well as D3. Regulation thresholds at the jack: x10 14.06 V typical / 14.86 V guaranteed, x100 14.42 / 15.22 V.

Thermal

Simulated: 3-D finite difference at 0.5 mm on the rasterised copper of all four layers (`thermal.py`), JLC 4-layer stack, h = 10 W/m2K per face in free air, 5 W/m2K to the case interior plus the case to ambient at 12 W/m2K over its outer area, rails at 1500 W/m2K contact. Validation: a uniformly heated plate lands 2.4 % from P/(2hA) and every solve balances its input power. The 10 A leads are modelled as fins (2.5 mm2 copper, 3 mm OD) where stated; without them the shunt island has no path but convection.

Load case	Enclosure	Ta C	P W	Case C	U6 C	R5 C	U2 C	PSU1 C	Rel1 C	U1 C
15 V x100, 10 A	open	25.0	2.36	25.0	77.4	192.7	148.5	151.2	58.1	70.3
15 V x100, 10 A	open	40.0	2.36	40.0	92.4	207.7	163.5	166.2	73.1	85.3
15 V x100, 10 A	case	25.0	2.36	42.9	94.0	311.7	265.6	268.3	70.6	86.9
15 V x100, 10 A	case	40.0	2.36	57.9	109.0	326.7	280.6	283.3	85.6	101.9
15 V x100, 10 A	open+leads	25.0	2.36	25.0	76.5	110.0	67.9	69.8	57.2	69.5
15 V x100, 10 A	open+leads	40.0	2.36	40.0	91.5	125.0	82.9	84.8	72.2	84.5
15 V x100, 10 A	case+leads	25.0	2.36	42.9	93.2	123.1	80.9	87.1	69.7	86.1
15 V x100, 10 A	case+leads	40.0	2.36	57.9	108.2	138.1	95.9	102.1	84.7	101.1
16 V x10, 10 A	open	25.0	2.17	25.0	76.8	192.5	148.2	151.0	48.8	68.7
16 V x10, 10 A	open	40.0	2.17	40.0	91.8	207.5	163.2	166.0	63.8	83.7
16 V x10, 10 A	case	25.0	2.17	41.4	92.9	310.1	263.9	266.6	62.3	84.8
16 V x10, 10 A	case	40.0	2.17	56.4	107.9	325.1	278.9	281.6	77.3	99.8
16 V x10, 10 A	open+leads	25.0	2.17	25.0	76.0	110.0	67.7	69.0	48.0	67.9
16 V x10, 10 A	open+leads	40.0	2.17	40.0	91.0	125.0	82.7	84.0	63.0	82.9
16 V x10, 10 A	case+leads	25.0	2.17	41.4	92.1	122.7	80.4	84.4	61.4	84.0
16 V x10, 10 A	case+leads	40.0	2.17	56.4	107.1	137.7	95.4	99.4	76.4	99.0
15 V x100, 0 A	open	25.0	1.36	25.0	76.3	44.4	55.1	69.5	56.9	69.2
15 V x100, 0 A	open	40.0	1.36	40.0	91.3	59.4	70.1	84.5	71.9	84.2
15 V x100, 0 A	case	25.0	1.36	35.4	85.6	68.5	67.9	79.5	62.1	78.5
15 V x100, 0 A	case	40.0	1.36	50.4	100.6	83.5	82.9	94.5	77.1	93.5



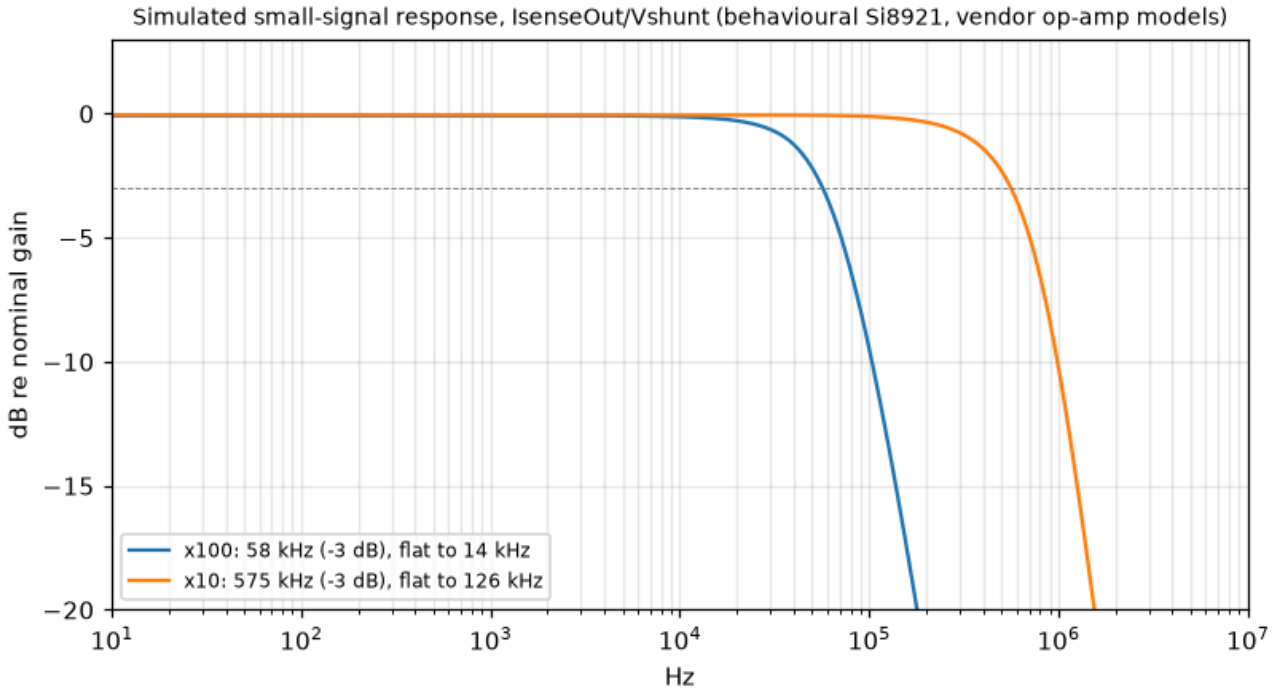
Outer-face temperature, 16 V on x10 with 10 A through the shunt, aluminium case at 40 C ambient, leads as fins (simulated).

With the board file's own stackup (1.524 mm core under the top layer) the worst-case board maximum rise is 159 C instead of 98 C, which is why the fabricated stack matters and is listed as a documentation item.

Analog: bandwidth, bias, noise, loop stability, step

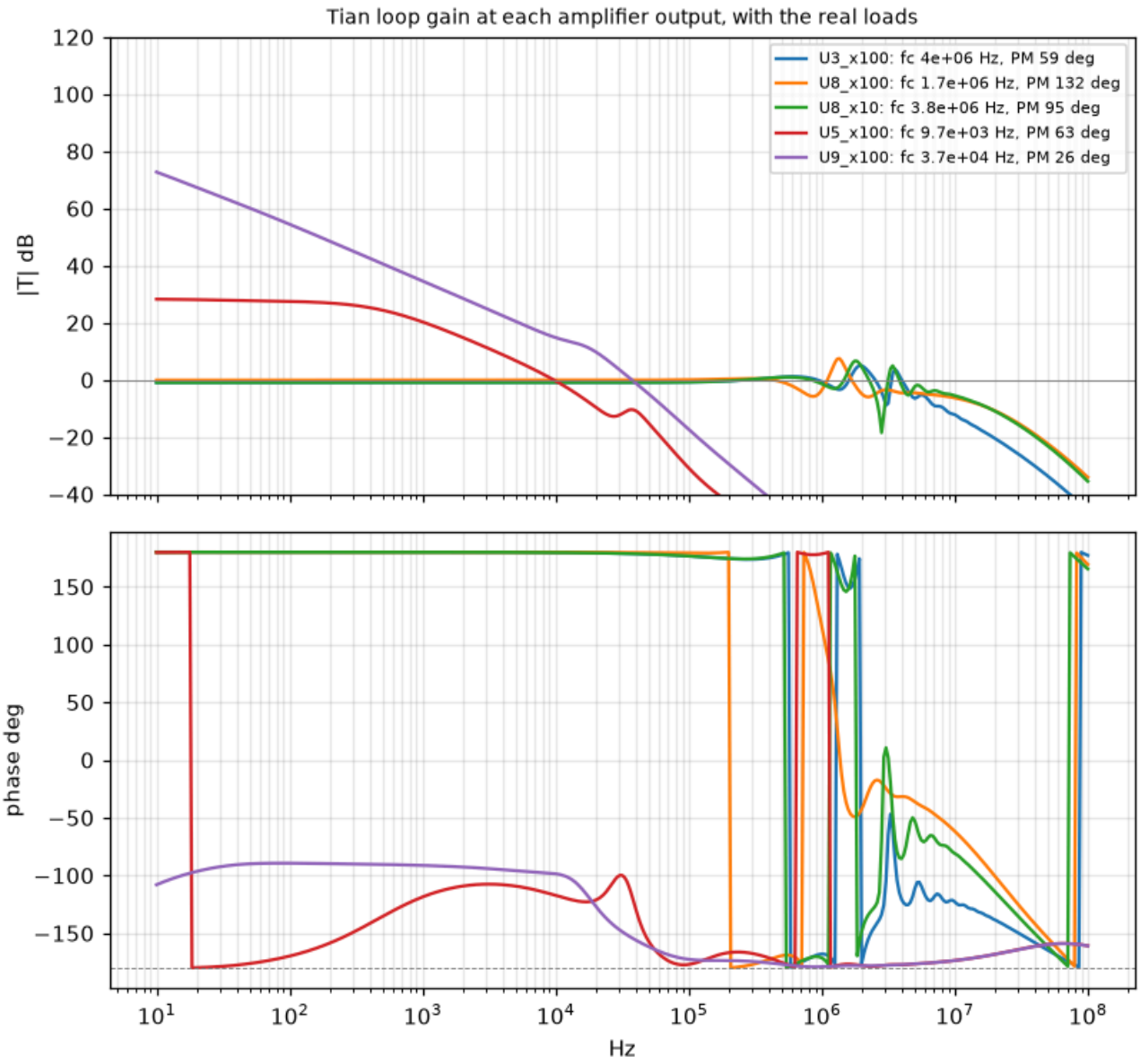
Simulated with the vendor OPA197 and LM7321 models (validated at 12 V in `sim/validate_models.py`: OPA197 Vos 19 uV, GBW 10.1 MHz, Iq 1.60 mA against 25 uV / 10 MHz / 1.0 mA typical; LM7321 10 V group Vos -1.72 mV, Ib -1.12 uA, GBW 9.1 MHz against +/-0.35 mV / 0.4 uA / 24 MHz - the model's bandwidth at this supply is low and its output swing optimistic, so the measured 0.30 V clamp is used for headroom) and a behavioural Si8921 from datasheet Table 5 (behavioural, Table 5: gain 8.2, two real poles (-3 dB 600 kHz), delay 0.80 us total, Vocm 1.39 V, clip 2.5 V diff, noise 79 nV/rtHz). Capacitors on the references are DC-bias derated (C30 17 uF, C29 13 uF).

Range	Gain	-3 dB kHz (sim)	-3 dB kHz (bench)	0.1 dB kHz	1 dB kHz
x100	99.14	57.5	60	14.5	38.0
x10	9.94	575.4	580	125.9	354.8



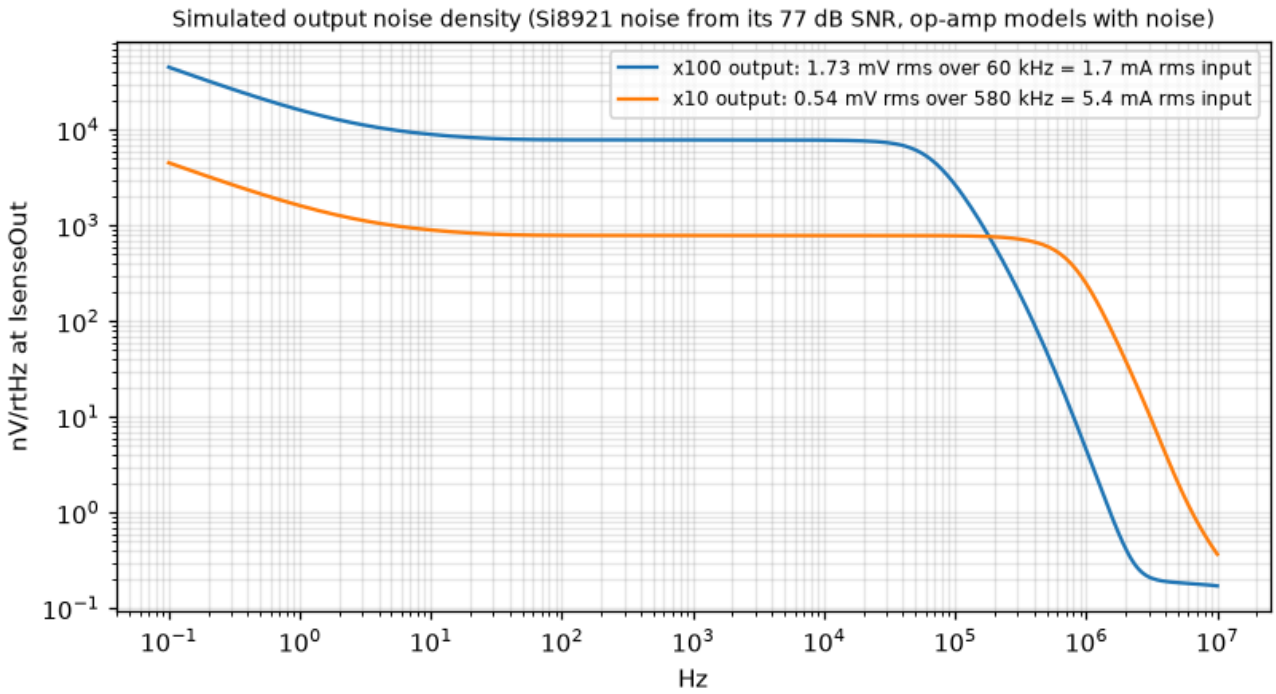
Small-signal response of the chain for both ranges (simulated).

Loop	Crossover Hz	Phase margin deg	Gain margin dB
U3_x100	3.99e+06	59.1	42.8
U8_x100	1.68e+06	132.2	29.9
U8_x10	3.82e+06	95.5	29.0
U5_x100	9.68e+03	63.2	60.9
U9_x100	3.74e+04	25.9	-



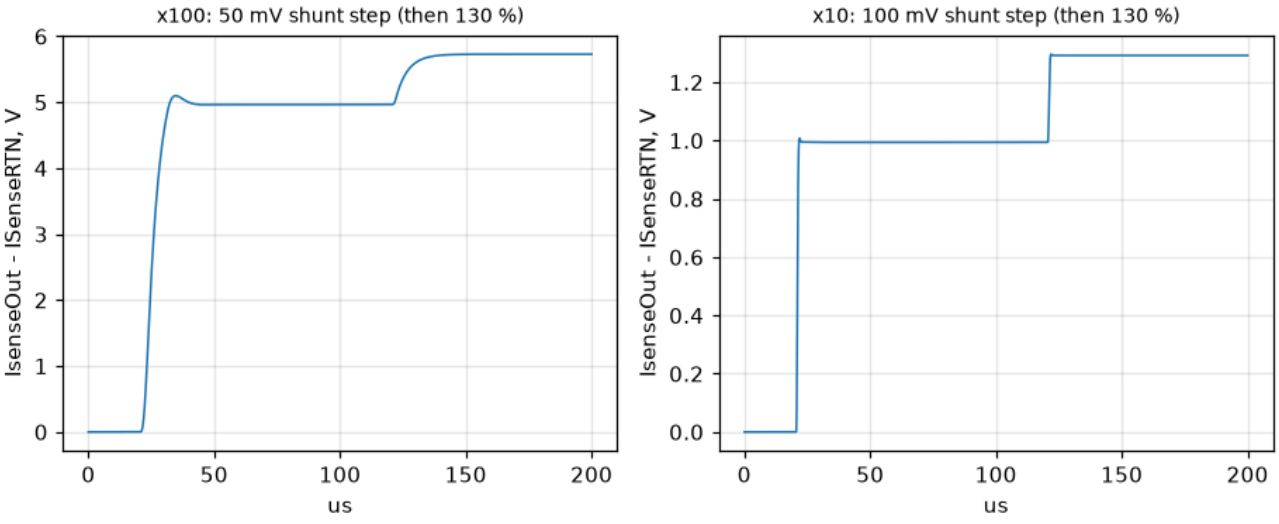
Tian loop gain at each amplifier output with the board's loads (simulated).

Range	Output density at 1 kHz nV/rtHz	Output rms over the band mV	Input-referred mA rms	Band kHz
x100	7885.9	1.732	1.73	60.0
x10	790.5	0.544	5.44	580.0



Output noise density (simulated); the Si8921's share is its 77 dB SNR as a white source.

Range	Step mV	Final V	Rise 10-90 us	Delay us	Overshoot %	Settle 0.1 % us	130 % step V
x100	50.0	4.965	7.15	5.17	2.70	22.7	5.729
x10	100.0	0.994	0.60	1.09	1.45	2.9	1.292



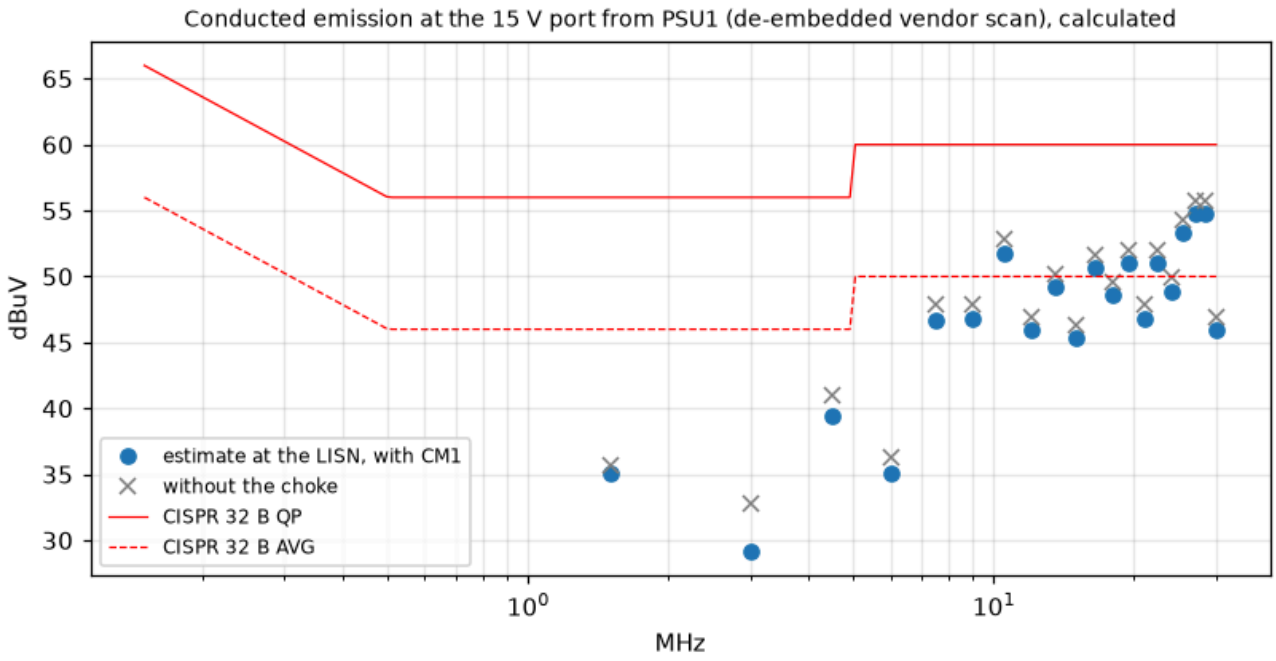
Full-scale step on each range, then 130 % of full scale (simulated; the clamp level reflects the model's optimistic swing).

+12 V	Range	Vshunt V	R21	ISenseRTN V	DiffAmpRef-ISenseRTN mV	U3 out V	U8 out V	Output V	I(+12) mA
12.0	x100	+0.000	0	6.054	-0.05	6.054	6.054	+0.0001	5.71
12.0	x100	+0.000	10000	6.054	+11.08	6.065	6.166	+0.1113	5.80
12.0	x100	+0.050	0	6.054	-0.30	6.551	11.019	+4.9642	9.46
12.0	x100	-0.050	0	6.054	+0.19	5.557	1.090	-4.9640	8.20
12.0	x10	+0.000	0	6.054	-0.05	6.054	6.054	+0.0000	5.71
12.0	x10	+0.000	10000	6.054	+11.11	6.065	6.065	+0.0112	5.72
12.0	x10	+0.100	0	6.054	-0.05	7.048	7.048	+0.9939	5.97
12.0	x10	-0.100	0	6.054	-0.06	5.060	5.060	-0.9939	5.46
10.6	x100	+0.000	0	5.354	-0.05	5.354	5.354	+0.0001	5.46
10.6	x100	+0.000	10000	5.354	+11.09	5.365	5.466	+0.1114	5.54
10.6	x100	+0.050	0	5.354	-0.29	5.851	10.319	+4.9643	9.20
10.6	x100	-0.050	0	5.354	+0.20	4.857	0.390	-4.9640	8.01
10.6	x10	+0.000	0	5.354	-0.05	5.354	5.354	+0.0000	5.46
10.6	x10	+0.000	10000	5.354	+11.12	5.365	5.365	+0.0112	5.46
10.6	x10	+0.100	0	5.354	-0.04	6.348	6.348	+0.9939	5.71
10.6	x10	-0.100	0	5.354	-0.05	4.360	4.360	-0.9939	5.20

Bias (**simulated**): at the 10.6 V rail (13 V in on x100) every amplifier still sits inside its input range; what runs out is the output swing at full scale, as the rails table shows. The DiffAmpRef-ISenseRTN difference is the LM7321 model's offset and bias current acting on the two dividers, and it doubles with the board's 10 k R21.

EMC

Calculated, module only (`emc_conducted.py`): PSU1's harmonics as de-embedded from Murata's conducted scan in the FTCAN review (same module, 1.5 MHz), as common-mode current through the 2.5 pF barrier, against CISPR 32 Class B DC-port limits. CM1 (1 mH) attenuates -0.6 dB at 1.5 MHz against a 40 kohm source. Worst margin -4.8 dB to the average limit. The Si8921 modulator has no vendor scan and the measured circuit's earth path is unknown; a LISN scan decides.



Conducted emission estimate at the 15 V port (calculated).

DFM and the fabrication package

Checked (`dfm_check.py`): narrowest copper {'1': 0.25, '32': 0.254} mm per layer against a 0.130 mm rule; closest different-net copper {'Top': '0.200 (NetR10_1)', 'Signal 1': '0.206 (NetL3_2)', 'Signal 2': '0.206 (NetL3_2)', 'Bottom': '0.206 (NetL3_2)'}; copper to the slot wall 0.21 mm on every layer against JLC's 0.30 mm routed-edge figure; annular rings at or above 0.15 mm (vias 0.6/0.3); the drill `.TXT` reconciles hole for hole (160 round holes plus 3 slots against 163 on the board); no fiducials, no test points.

Sev	Finding
2	Pours come within 0.21 mm of the barrier slot wall on every layer
2	Barrier slot is drawn on the Outline layer as a 2.5 mm stroked path, not a closed cut-out contour
2	A .GKO (keep-out) file with an 11.5 x 3.9 mm rectangle is in the fabrication set
3	No fiducials and no test points on the board
3	J2-to-shunt pours rise ~21 C at 10 A (IPC-2221, 1 oz, top only)

Enclosure

Calculated (`enclosure_fit.py`): the board's mid-plane sits 8.75 mm above the panel bottom (the SMA cutout), bottom side up. With 1.5 mm walls (assumed; the case has no drawing) there is 12.95 mm above the bottom side and 6.45 mm above the top side.

Part	Side	Height mm	Margin mm	Source
C7 standing	BOTTOM	16.0	-3.05	a750.pdf: MV size code = 10 x 16 mm can; if mounted upright
J4	BOTTOM	11.0	+1.95	pj-002a.pdf drawing, 11.00
J2	BOTTOM	10.5	+2.45	EST: BCH-508HS 2-pos header, through Panel B cutout 9.5 mm tall
C7 lying (RA1)	BOTTOM	10.0	+2.95	a750.pdf: 10 mm diameter; the footprint name (RA1) implies a right-angle, lying mount
U2	TOP	2.8	+3.65	board file HEIGHT 2.8
D4/D5	BOTTOM	9.0	+3.95	panel LED holes at Y 18.0: leads formed to reach the panel, EST
Q5/Q6	TOP	2.4	+4.05	board file HEIGHT 2.38
U1/U6	TOP	2.0	+4.45	board file HEIGHT 2.0
C11/C12	TOP	1.6	+4.85	EST: 1206 4.7 uF 50 V
C1 (0805 HD)	TOP	1.4	+5.05	board file HEIGHT 1.4
CM1	BOTTOM	5.0	+7.95	board file HEIGHT 5.0
R28	BOTTOM	5.0	+7.95	EST: CT-94 top-adjust body
SW1	BOTTOM	4.7	+8.25	os103011.pdf: frame H = 4.7 mm; actuator passes through the panel slot
Rel1	BOTTOM	4.2	+8.75	g6l.pdf G6L-1F drawing 4.2 +/-0.2
PSU1	BOTTOM	4.0	+8.95	board file HEIGHT 4.0

Sourcing and lifecycle (web check 2026-10-29)

Ref	Part	Status	Source	Stock / price
U5, U9	LM7321MF/NOPB	OBSOLETE	ti.com part details 2026-10-29: 'no longer in production'; Digi-Key: 'Obsolete and no longer manufactured'	TI none; Digi-Key none; LCSC 76 'discontinued' / None
Rel1	G6L-1F DC12	conflicting	Omron KR snippet 'In production'; Digi-Key CA 'Active' but no backorders, 'temporary constrained supply'; Mast	Digi-Key 0; Mouser ~5,220 (snippet); Master 639 / Mouser ~$1.78 (snippet)
J4	PJ-002A	conflicting	Master Electronics 'no longer being manufactured ... End of Life' (template wording); Same Sky page did not re	Master 31,656; Future 24,730 / Master $0.412 @100
CM1	DR331-105BE	conflicting	Digi-Key CA 'Active', non-cancellable; Master Electronics 'no longer being manufactured' (template wording); O	Digi-Key 0 (1,500 est. 2027-04-15, 16 wk); Master 4,887 / Master $0.5085 @500
C11, C12	CL31B475KBHNNNE	NRND (LCSC)	LCSC snippet 'Not recommended for new designs'; Samsung page HTTP 500	LCSC 86,430; Future 380,000 (40 wk) / LCSC from $0.0188

Every other line read Active at a distributor or could not be verified; the full table is `hardware/review/sourcing.json`.

The 2026-08-20 findings, re-examined

Id	Status	Finding	This run
F1	closed	Isolated auxiliary supply requirement was undocumented	ISenseRTN is bonded to chassis and board GND floats 6 V below it; an earthed supply shorts U5. README, CONNECTORS.md and the Rev 3.1 silkscreen now state it. Th
F2	open	+12 rail unregulated below ~14.9 V at the jack	UA78L12 needs 14.5 V in; one Schottky on the x10 range. Headroom on the 5 A range shrinks to 0.32 V at 13 V. Re-examined in this review with both diode drops on
F3	open	R21 asymmetry between the two reference buffers (10 k vs 0)	Recommended 100 ohm. The sheets now carry R21 = 0 ohm (CR0805-J/-000ELF); the board still carries CRCW080510K0FKTA (10 k) - see sync-1.
F4	open	R28 is 100 ohm on the schematic and CT94EW103 (10 k) in the BOM	A 10 k trimmer resolves to 5.7 % of full scale on the x100 range. BOM line unchanged in the 2026-08-20 Rev 3 output.
F5	open	U6 HT7550-1 dissipates ~0.55-0.6 W in SOT-89	Drops VregIn to 5 V for the Si8921 output side and the DC-DC module. Total board draw measured 95 mA; U6 share and junction temperature not measured. Re-examine
F6	closed	Published common-mode figure exceeded the J2 400 V connector rating	Corrected to 400 Vdc in the documentation; creepage and clearance were never checked on the layout. Re-examined in this review.
F7	closed	Accuracy stack-up: +/-(1.0 % + 12 mA) and +/-(0.9 % + 8 mA)	Hybrid worst case; bench 2026-09-06 measured -0.24 % and -0.05 % gain error on one board.
F8	closed	x100 bandwidth not derivable from the schematic	Measured 60 kHz on Rev 3.1. This review derives it: 50 ohm (R4
F9	open	C16, C17, C28 at 3 pF are below the layout parasitics	Set nothing repeatable; raise or delete.
F10	open	Difference-amp CMRR relies on four discrete resistors	R2/R9 8.2 k 0.05 % and R3/R20 10 k 0.1 %; a matched network would remove the drift term.
F11	open	No reverse-polarity, fuse or TVS on the 15 V input; D6 clamp topology	C7 (1000 uF polymer) reverse-biased by a centre-negative adapter. D6 is an antiparallel pair across AIP/AIN (not a series pair) - see protection-1.
F12	open	SW1 carries the whole board supply at its 100 mA / 12 V rating	Confirmed from the netlist in this review: pole pin 2 fed by D3, positions 3 and 4 carry VregIn and HighGain. 95 mA measured at 15 V.
F13	open	BOM lines C8, C43, R23 are generic placeholders	680 pF, 680 pF and 1 ohm on the sheet; no part number in the 2026-08-20 Rev 3 BOM.
F14	open	Relay contact resistance sits in the x100 gain network	~100 mohm in series with 156.7 ohm: 0.06 % initially, a wear-out term.
F15	open	Rev 3.1 silkscreen legends disagree with the schematic	x50 fixed to x100 (2026-08-21); Vout settled at 5.70 V measured. Still open: Supply 13-16Vdc against the 12 V relay coil (15.6 V max), Max Input +/-200mV agains

Inputs, models and what was assumed

Datasheets read in this session are in `hardware/review/datasheets/` (24 files; `ds_params.py` names the table or page behind each number). Models: TI OPA197.LIB (2019), the National LM7321.MOD with its 10 V group active, a behavioural Si8921, no model for the UA78L12 (TI publishes only the 5 V variant) or the HT7550-1. Assumed: IEC 60664-1 figures at PD2 / group IIIb / OVC II; the JLC stackup; 1.5 mm case walls and 1 mm rail engagement; 2.5 mm2 leads; X5R DC-bias derating from a fitted class-2 curve; the module's harmonics from Murata's scan; h = 10 / 5 / 12 W/m2K.

Not assessable here: the real shunt temperature at 10 A (needs a measurement with the leads used), the conducted scan, the creepage clause the product will claim, the case section (wall thickness, groove depth).